Towards Fluid Networking: Citizen-Co-Created Low-Height Distributed MIMO Access Networks

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***Abstract*—This paper studies low-height distributed-multiple-input multiple-output (MIMO) access networks with repositionable access points in an outdoor urban setting. Using the `COCOON-Sionna` framework, the workflow generates wall-mounted candidate access points (APs), simulates pedestrian and bicycle mobility, computes path-based channel state information (CSI), and compares a central rooftop proxy, a fixed distributed deployment, and two proxy-based relocation-capable distributed deployments over a common scenario. The article formalizes the signal-to-interference-plus-noise ratio (SINR)- and ergodic sum rate (ESR)-based evaluation model, a nearest-user user equipment (UE)-to-UE proxy-CSI heuristic that evaluates a proxy-ESR objective from wideband proxy channels, and a nearest-user proxy-power heuristic that ranks candidate subsets from stored UE-to-UE link-power measurements scaled to the AP transmit-power budget. In the Rabot scenario, the proxy-ESR controller is the strongest strategy: it reaches a mean ESR of 133.1 bit/s/Hz, improves the 10th-percentile ESR by 88.8 % over the fixed distributed baseline, and reduces the 0 dB outage fraction from 4.41 % to 1.87 %. The power-only proxy controller is second-best in radio performance but requires more motion. The paper also documents the postprocessing workflow that produces geometry, performance, and relocation-cost figures from reproducible simulation outputs.**

***Index Terms*—cell-free systems, distributed MIMO, low-height access points, access-point placement, urban mobility, Sionna RT**

I. Introduction

Wireless access networks have traditionally been designed under the assumption of static infrastructure and slowly varying demand. Even in advanced multi-antenna systems such as massive MIMO [1], [2] and cell-free (CF) architectures [3], [4], the physical deployment of access points (access points (APs)) is typically fixed, while adaptability is achieved through signal processing, scheduling, and beamforming. Although these approaches significantly improve spectral efficiency and coverage, they inherently assume that spatial mismatches between infrastructure and users can be mitigated purely at the signal level.

Recent research has shown that bringing antennas closer to users through distributed deployments improves both energy and spectral efficiency, but also makes performance strongly dependent on the actual placement of the distributed infrastructure [5]–[7]. This shifts part of the design problem from signal processing to geometry. In parallel, realistic channel modeling and measurement campaigns have enabled more faithful evaluation of wireless systems in dynamic environments. Geometry-based stochastic channel models such as QuaDRiGa [8] and measurement-driven studies in mobile and vehicular scenarios [9]–[11] provide detailed insight into spatial and temporal channel dynamics. More recently, differentiable and site-specific ray-tracing tools such as Sionna RT have made it possible to evaluate radio systems directly in detailed three-dimensional scenes, bridging the gap between abstract channel models and environment-aware wireless design [12]. However, despite these advances, most existing works still treat infrastructure placement as either static or offline-optimized, rather than as a variable that can be repeatedly adapted over time.

At the same time, the evolution toward 6G systems emphasizes extreme densification, distributed intelligence, and tight integration between communication, sensing, and control [13]–[15]. Concepts such as cell-free networking, radio stripes, and distributed antenna systems aim to provide ubiquitous service by leveraging spatial diversity and user-centric operation [16], [17]. As this line of work matures, mobility is becoming a central challenge. In particular, recent work on mobile cell-free massive MIMO highlights that user mobility fundamentally affects channel acquisition, coordination, scalability, and practical deployment decisions, and therefore requires new architectures and control strategies beyond static infrastructure assumptions [18]. Yet, even in that context, adaptation is still predominantly considered at the logical or signal-processing level, rather than through physically reconfigurable low-height infrastructure.

The Cocoon project departs from these conventions by introducing a fluid networking paradigm, in which infrastructure itself becomes part of the adaptive loop. Instead of optimizing only transmission strategies, the network can reconfigure its physical topology by repositioning low-height APs based on measured channel conditions. This concept builds on recent advances in experimental testbeds and open frameworks for 6G research [19]–[21], while extending the state of the art toward infrastructure-aware adaptation in realistic urban environments.

The Cocoon project targets a wireless access paradigm in which the network adapts to the fluidity of the city rather than assuming a static infrastructure and a stationary demand profile. The motivating hypothesis from the project proposal is that a practical CF network becomes more feasible when low-height infrastructure can be co-created, repositioned, and activated only when the measured channel conditions justify it. In that view, the citizens are not only traffic sources but

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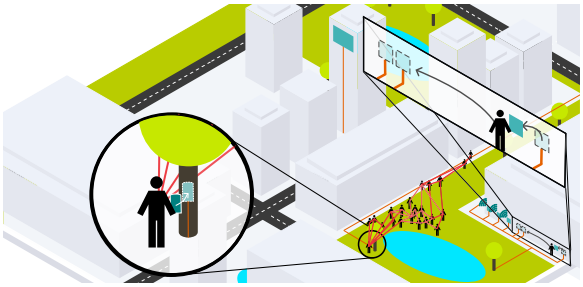


Fig. 1: Conceptual illustration of the Cocoon low-height access-network vision, where citizens interact with the network and reposition APs according to measured channel conditions and user locations. The links between people represent real-time channel-gain measurements; the person close to the tree illustrates a favorable candidate location for the next low-height AP.

also anchors for sensing, mounting, and reconfiguring the surrounding network.

This article narrows that broader vision to a simulation-based study of low-height infrastructure adaptation. The implemented framework covers wall-mounted candidate APs, outdoor user mobility, path-based channel state information (CSI) extraction, and repeated placement decisions over relocation windows in an urban scene. By combining distributed-MIMO evaluation with site-specific ray tracing and mobility-aware infrastructure reselection, the work positions itself at the intersection of adaptive cell-free networking, realistic radio-environment modeling, and practical AP placement research [7], [12], [18]. This scope enables a reproducible comparison of geometry, radio-performance, and relocation-cost metrics for alternative infrastructure strategies. :contentReference[oaicite:0]index=0

Within this scope, the work focuses on evaluating how infrastructure mobility interacts with channel-aware decision-making. In particular, it considers distributed low-height deployments, a centralized rooftop proxy, and relocation-capable strategies that exploit proxy-based channel knowledge to guide AP placement. The emphasis is not on full-stack network operation, but on isolating the geometric and physical-layer effects of adaptive infrastructure under realistic mobility and propagation conditions.

The main contributions of the present study are:

- a concise technical framing of Cocoon as a low-height, movable-AP, citizen-co-created access-network concept,
- a reproducible description of the Rabot outdoor simulation pipeline implemented in `COCOON-Sionna`, and
- a common analysis workflow for geometry, signal-to-interference-plus-noise ratio (SINR), ergodic sum rate (ESR), and relocation-cost results derived from exported simulation artifacts.

II. Scope Within Cocoon

The full Cocoon proposal spans citizen interaction with the wireless network, channel assessment, architecture design, and proof-of-concept validation. The present article addresses the simulation-facing part of that program, with emphasis on dynamic channel-aware adaptation, low-height AP/contact service point (CSP) placement, and the algorithmic consequences of allowing physical repositioning.

The main modeling simplification is that co-creation is represented here through an abstract movable infrastructure layer rather than through a user study or a deployment trial. Candidate mounting locations stand in for positions that could later be offered to citizens, and the placement scheduler stands in for the future tool chain that would notify participants where the next best CSP constellation should be installed.

Accordingly, the scope of the present article is:

- the paper validates a simulation methodology and a comparison workflow, not the social feasibility of citizen participation,
- the paper studies a Rabot outdoor scenario, not yet a multi-site or multi-season campaign,
- the paper compares placement strategies over a common CSI-derived objective, not yet a full end-to-end network stack.

A. Placement Strategies

The placement layer compares the following strategy definitions under a common AP budget:

1 Central massive MIMO – one rooftop proxy is placed on the building whose rooftop representative point is nearest the area center and is oriented toward that center. Its antenna topology is the normalized co-located 6×6 uniform planar array (UPA) obtained by aggregating the distributed element budget into one rooftop site; it is used only as the central reference.

2 Distributed fixed – one distributed subset is sampled from the wall-mounted candidate pool and kept unchanged over the full trajectory.

3 Distributed movable (Opt. 1) – optimization 1. The distributed subset is reselected every relocation window from a nearest-user user equipment (UE)-to-UE proxy-CSI model. Each candidate site binds to the closest user within d_{th}, the controller forms a wideband proxy channel from that user's measured UE-to-UE CFR, and the subset is ranked by proxy ESR.

4 Distributed movable (Opt. 2) – optimization 2. The distributed subset is reselected every relocation window from the same nearest-user distance-threshold proxy model, but the controller uses the stored UE-to-UE link-power row of the anchor user, replaces the diagonal entry by the strongest non-self value, rescales by $P_{\mathrm{AP}}/P_{\mathrm{UE}}$, and ranks the subset by proxy average power.

III. System Model

We consider a downlink distributed MIMO (D-MIMO) system with active APs $\mathcal{A} = \{1, \ldots, M\}$ and scheduled UEs $\mathcal{U} = \{1, \ldots, K\}$. Let N_m denote the number of transmit ports at AP m, let $N_{r,k}$ be the number of receive ports at UE k, and let $f \in \{1, \ldots, F\}$ index the sampled CFR tones over the 100 MHz band. The per-AP channel block on tone f is

$$\mathbf{H}_{k,m}[f] \in \mathbb{C}^{N_{r,k} \times N_m}, \tag{1}$$

and the full distributed channel observed by UE k is

$$\mathbf{H}_k[f] = [\mathbf{H}_{k,1}[f]\ \mathbf{H}_{k,2}[f]\ \cdots\ \mathbf{H}_{k,M}[f]]\,. \tag{2}$$

The antenna topology follows the `PlanarArray` abstraction used by the simulator. Each node is modeled as a UPA with N^{row} rows and N^{col} columns, so that the number of polarization-agnostic elements is $N^{\mathrm{row}}N^{\mathrm{col}}$. The implementation uses the same element radiation pattern, polarization setting, and inter-element spacing on transmit and receive arrays. More precisely, the number of ports is

$$N = N^{\mathrm{row}}N^{\mathrm{col}}N^{\mathrm{pol}}, \tag{3}$$

where N^{pol} is the number of active polarization ports implied by the configured polarization string. In the Rabot configuration, the APs use a 2×3 UPA with `VH` dual polarization, which yields $N_m = 2\cdot3\cdot2 = 12$ transmit ports per AP. The UEs use a 1×1 planar array with the same `VH` dual-polarized element model, which yields $N_{r,k} = 1\cdot1\cdot2 = 2$ receive ports per user device. The vertical and horizontal element spacings are both set to 0.5λ, so the array factor corresponds to half-wavelength sampling in both planar directions.

The element radiation pattern is the 3rd generation partnership project (3GPP) `tr38901` pattern provided by Sionna. Thus, the simulated channel is not isotropic at the element level: every array response combines the `tr38901` element directivity with the UPA array factor determined by the node orientation, row-column topology, and the propagation geometry. This matters for the present study because the wall-mounted distributed APs and the rooftop central proxy are evaluated with the same underlying element pattern while differing in their locations, orientations, and distributed versus centralized array geometry.

For the central reference, the implementation collapses the distributed element budget into one normalized co-located rooftop array. If each distributed AP has $N_{\mathrm{AP}}^{\mathrm{row}}N_{\mathrm{AP}}^{\mathrm{col}}$ physical element locations and the distributed budget contains M_{dist} APs, then the central proxy uses

$$N_{\mathrm{cent}}^{\mathrm{elem}} = M_{\mathrm{dist}}N_{\mathrm{AP}}^{\mathrm{row}}N_{\mathrm{AP}}^{\mathrm{col}} \tag{4}$$

element locations, reshaped into a factorized UPA with integer row and column counts. In the Rabot configuration, $M_{\mathrm{dist}} = 6$ and the distributed hardware uses 2×3 elements per AP, so the normalized co-located rooftop proxy becomes a 6×6 UPA. With the same `VH` dual polarization, this yields $6\cdot6\cdot2 = 72$ transmit ports. The central transmit-power budget is scaled to the sum power of the 6 distributed APs. This keeps the central reference aligned with the distributed element and power budgets, while still remaining a rooftop proxy rather than a full placement model for a true co-located base-station site.

Given a receive combiner $\mathbf{u}_k[f]$, the effective channel becomes

$$\mathbf{h}_k^{\mathsf{H}}[f] = \mathbf{u}_k^{\mathsf{H}}[f]\mathbf{H}_k[f]. \tag{5}$$

Stacking all users yields the effective distributed channel matrix $\mathbf{H}_{\mathrm{eff}}[f]$. The current pipeline applies a least-squares pseudo-inverse zero-forcing (ZF) precoder independently on every sampled tone, followed by per-stream power allocation under a total AP power budget.

For UE k on tone f, the instantaneous downlink SINR is evaluated as

$$\mathrm{SINR}_k^{\mathrm{DL}}[f] = \frac{p_k\left|\mathbf{h}_k^{\mathsf{H}}[f]\mathbf{f}_k[f]\right|^2}{\sum_{i\in\mathcal{U},\,i\neq k} p_i\left|\mathbf{h}_k^{\mathsf{H}}[f]\mathbf{f}_i[f]\right|^2 + \sigma^2}, \tag{6}$$

where $\mathbf{f}_i$ is the precoding vector for stream i, p_i its allocated power, and σ^2 the thermal-noise term. The implementation keeps the interference term explicitly and exports desired-signal, interference, and noise power terms for each evaluated user snapshot.

The implementation evaluates this expression on all sampled tones and stores a scalar wideband SINR per user by mapping the tone-averaged spectral efficiency back to a Shannon-equivalent SINR. The ESR used later in the results section is derived from the same full-band per-snapshot SINR samples. For snapshot n with scheduled-user set $\mathcal{U}_n$, the postprocessing layer computes

$$\mathrm{ESR}_n = \sum_{k\in\mathcal{U}_n}\frac{1}{F}\sum_{f=1}^{F}\log_2\!\left(1+\mathrm{SINR}_{k,n}^{\mathrm{DL}}[f]\right). \tag{7}$$

These per-snapshot SINR values form the basis of the trajectory-level SINR and ESR statistics reported later in the article.

The placement strategies compared throughout the article are defined in Section II-A. Let $\mathcal{C}$ denote the wall-mounted candidate pool and let $\mathcal{S}_w \subseteq \mathcal{C}$ denote the selected movable-AP subset in relocation window w.

IV. Relocation Heuristic

The relocation mechanism repeats the same observe-score-relocate loop at every relocation instant. The Distributed movable (Opt. 1) and Distributed movable (Opt. 2) rules assume that the AP-to-UE channel of a dormant candidate position is not known before an AP is actually moved there. Consequently, these two rules do not optimize directly over candidate-position AP-to-UE CSI. Instead, each candidate site is associated with the closest user within a distance threshold. The Distributed movable (Opt. 1) rule reuses the measured UE-to-UE wideband channel of that nearby user as a proxy and ranks the subset by proxy ESR. The Distributed movable (Opt. 2) rule uses the corresponding stored UE-to-UE link-power samples, scaled to the AP transmit-power budget, and ranks the subset by proxy average power. In the present trace-driven simulation, the selector is given the user positions and UE-to-UE proxy measurements over the current relocation window as a short-horizon look-ahead signal. This is future information relative to a strictly causal online controller, but it is not candidate-position AP-to-UE CSI. Once the best subset has been chosen, the movable APs are relocated to those candidate positions and the actual deployed AP-to-UE CSI is recomputed for storage, postprocessing, and final reporting.

The movable distributed strategy is implemented as a windowed receding-horizon controller. Let $\{t_n\}_{n=1}^{N_t}$ denote the simulated snapshot times and let T_{rel} denote the relocation interval. The trajectory is partitioned into disjoint decision windows

$$\mathcal{W}_w = \left\{ n \in \{1, \ldots, N_t\} : \left\lfloor \frac{t_n - t_1}{T_{\mathrm{rel}}} \right\rfloor = w - 1 \right\}. \tag{8}$$

At the beginning of each window w, the controller selects a subset $\mathcal{S}_w \subseteq \mathcal{C}$ with cardinality $|\mathcal{S}_w| = N_{\mathrm{mov}}$, where N_{mov} is the movable-AP budget. The fixed distributed mode uses one time-invariant subset, whereas the movable mode recomputes $\mathcal{S}_w$ at every relocation instant. A strictly causal online variant would replace the current-window samples below by only the decision-time sample, or by predicted samples for the upcoming window.

A. Distributed movable (Opt. 1) – current-window nearest-user proxy-CSI controller.

Let d_{th} denote the configured distance threshold, let $\mathbf{r}_{u,n}$ be the horizontal position of user u at snapshot n, and let $\mathbf{x}_c$ be the horizontal coordinate of candidate site c. For each candidate site $c \in \mathcal{C}$ and snapshot $n \in \mathcal{W}_w$, define the anchor user

$$\ell(c, n) = \arg\min_{u \in \mathcal{U}} \|\mathbf{r}_{u,n} - \mathbf{x}_c\|_2 , \tag{9}$$

and the validity indicator

$$\delta(c, n) = \mathbf{1}\left\{ \left\|\mathbf{r}_{\ell(c,n),n} - \mathbf{x}_c\right\|_2 \le d_{\mathrm{th}} \right\}. \tag{10}$$

The AP-to-UE CSI for a dormant candidate site is not available before relocation. Therefore, the Distributed movable (Opt. 1) rule replaces it by a proxy channel extracted from the measured UE-to-UE CSI. If $h^{\mathrm{UU}}_{u,v,n}[f]$ denotes the sampled UE-to-UE coefficient from transmitter-user v to receiver-user u at snapshot n and tone f, the proxy coefficient used for candidate c is

$$\tilde{h}_{u,c,n}[f] = \begin{cases} h^{\mathrm{UU}}_{u,\ell(c,n),n}[f], & \delta(c,n) = 1,\ u \neq \ell(c,n), \\ h^{\mathrm{UU}}_{v^\star,\ell(c,n),n}[f], & \delta(c,n) = 1,\ u = \ell(c,n), \\ 0, & \delta(c,n) = 0, \end{cases} \tag{11}$$

where $v^\star = \arg\max_{v \in \mathcal{U}} \max_f |h^{\mathrm{UU}}_{v,\ell(c,n),n}[f]|$ supplies a surrogate self-link because the diagonal user-to-itself measurement is not observable. For a candidate subset $\mathcal{S}$, the proxy channel matrix $\tilde{\mathbf{H}}_n[f](\mathcal{S})$ is formed by stacking the columns $\tilde{h}_{u,c,n}[f]$ for all $c \in \mathcal{S}$. The same ZF procedure as in the deployed AP case is then applied on every sampled tone, yielding a proxy SINR $\tilde{\gamma}_{u,n}[f](\mathcal{S})$ and the current-window proxy-ESR objective

$$J^{(1)}_w(\mathcal{S}) = \frac{1}{|\mathcal{W}_w| F} \sum_{n \in \mathcal{W}_w} \sum_{f=1}^{F} \sum_{u \in \mathcal{U}} \log_2(1 + \tilde{\gamma}_{u,n}[f](\mathcal{S})). \tag{12}$$

The summation over $n \in \mathcal{W}_w$ is therefore an average over trace samples in the relocation interval for which $\mathcal{S}_w$ remains deployed. It is a proxy-ESR score computed from UE-to-UE look-ahead measurements, not from candidate-position AP-to-UE CSI.

B. Distributed movable (Opt. 2) – current-window nearest-user proxy-power controller.

The proxy-power rule reuses the same nearest-user anchor $\ell(c, n)$ and validity indicator $\delta(c, n)$ as Distributed movable (Opt. 1), but it does not use the complex proxy coefficient directly. Instead, it operates on the stored UE-to-UE link-power tensor. Let $\rho^{\mathrm{UU}}_{a,u,n}$ denote the ordered link-power entry returned by the simulator between anchor-user index a and user index u at snapshot n. The implementation forms the per-candidate proxy-power row

$$\bar{\rho}_{u,c,n} = \begin{cases} \rho^{\mathrm{UU}}_{\ell(c,n),u,n}, & \delta(c,n) = 1,\ u \neq \ell(c,n), \\ \max_{v \in \mathcal{U},\, v \neq \ell(c,n)} \rho^{\mathrm{UU}}_{\ell(c,n),v,n}, & \delta(c,n) = 1,\ u = \ell(c,n), \\ 0, & \delta(c,n) = 0, \end{cases} \tag{13}$$

and then rescales that row to the distributed-AP transmit-power budget according to

$$\tilde{\rho}_{u,c,n} = \frac{P_{\mathrm{AP}}}{P_{\mathrm{UE}}} \bar{\rho}_{u,c,n}. \tag{14}$$

The Distributed movable (Opt. 2) objective is then

$$J^{(2)}_w(\mathcal{S}) = \frac{1}{|\mathcal{W}_w|} \sum_{n \in \mathcal{W}_w} \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \sum_{c \in \mathcal{S}} \tilde{\rho}_{u,c,n}. \tag{15}$$

Distributed movable (Opt. 2) is therefore the power-only counterpart of Distributed movable (Opt. 1): it uses the same nearest-user proxy association and the same self-link substitution rule, but it requires only stored proxy link power rather than the complex proxy channel.

C. AP Relocation Selection – Greedy subset construction.

Both proxy-based optimization rules use the same forward-selection search. Starting from $\mathcal{S}^{(0)}_w = \varnothing$, the controller adds one candidate at a time according to

$$c^\star_r = \arg\max_{c \in \mathcal{C} \setminus \mathcal{S}^{(r-1)}_w} J^{(q)}_w\left(\mathcal{S}^{(r-1)}_w \cup \{c\}\right), \tag{16}$$

$$q \in \{1, 2\}, \qquad \mathcal{S}^{(r)}_w = \mathcal{S}^{(r-1)}_w \cup \{c^\star_r\}, \qquad r = 1, \ldots, N_{\mathrm{mov}}, \tag{17}$$

and finally sets $\mathcal{S}_w = \mathcal{S}^{(N_{\mathrm{mov}})}_w$. This is a forward-selection heuristic rather than an exhaustive combinatorial optimizer.

D. Physically Relocating APs

After either optimization rule has chosen the subset, the selected geometric candidate sites are converted into physical relocations of labeled movable APs. Let $\mathbf{p}^{(w-1)}_m$ denote the previous-window position of movable AP m and let $\mathbf{q}^{(w)}_j$ denote the coordinates of the candidates in the newly selected subset. The controller solves the linear assignment problem

$$\pi^\star_w = \arg\min_{\pi \in \Pi_{N_{\mathrm{mov}}}} \sum_{m=1}^{N_{\mathrm{mov}}} \left\| \mathbf{p}^{(w-1)}_m - \mathbf{q}^{(w)}_{\pi(m)} \right\|_2^2, \tag{18}$$

where $\Pi_{N_{\mathrm{mov}}}$ is the permutation set on $\{1, \ldots, N_{\mathrm{mov}}\}$. This minimum-displacement matching preserves AP identities across windows and yields schedule traces that correspond

to physically meaningful AP trajectories instead of arbitrary candidate relabelings. This allows to investigate the required relocation distances, i.e., which strategies require the most relocations.

V. Simulation Setup

The Rabot outdoor scenario is defined in `scenarios/rabot.yaml`. The scene is derived from an OpenStreetMap (OSM) boundary around the KU Leuven Gent Campus Rabot environment. Candidate distributed-AP positions are generated along building walls at low height, one rooftop proxy is reserved for the central reference strategy, and user mobility is simulated over the outdoor walkable area with mixed pedestrian and bicycle profiles.

The simulation workflow used by the paper matches the repository workflow:

1) build the outdoor scene and walk graph from OSM,
2) generate wall-mounted candidate APs and the single rooftop proxy site,
3) simulate user trajectories over the walkable outdoor area,
4) compute path-based CSI and explicit SINR terms,
5) compare Central massive MIMO, Distributed fixed, Distributed movable (Opt. 1), and Distributed movable (Opt. 2),
6) export reusable result files,
7) generate analysis figures in a separate postprocessing step.

This section documents the reproducible processing chain from scene generation to postprocessed figure export. The antenna configuration is kept fixed across all placement strategies at the element-model level: the simulator instantiates UPA-based node arrays with the 3GPP `tr38901` element pattern, half-wavelength spacing in both planar dimensions, and `VH` dual polarization. The distributed strategies use identical 2×3 wall-mounted APs, whereas the central reference aggregates that same element budget into one normalized co-located rooftop 6×6 array with summed transmit power. Consequently, the comparison isolates the effect of geometry, relocation, and distributed-versus-co-located topology while keeping the underlying element directivity model unchanged.

The figure set in Sections VI to VIII is read directly from the generated Rabot outputs under `outputs/rabot/` and `outputs/rabot/postprocessing/`. This keeps the manuscript synchronized with the latest postprocessed result files in the repository.

For the current Rabot instantiation, the relocation controller starts from a single seed constellation drawn from the wall-mounted candidate pool using the configured random seed. That seed constellation is frozen for Distributed fixed and also initializes Distributed movable (Opt. 1) and Distributed movable (Opt. 2). Thereafter, every relocation window applies the greedy subset search from Section III with the selected proxy objective, followed by a minimum-displacement assignment that preserves persistent AP identities across windows. The actual deployed AP-to-UE CSI is recomputed only after the new candidate subset has been selected.

TABLE I: Key Rabot scenario parameters used in the Rabot simulations.

Parameter	Value
Carrier frequency f_c	3.5 GHz
Bandwidth B	100 MHz
CFR tones F	128
AP array topology $N_{\mathrm{AP}}^{\mathrm{row}} \times N_{\mathrm{AP}}^{\mathrm{col}}$	2×3 UPA, dual-polarized `VH`
Central proxy topology $N_{\mathrm{cent}}^{\mathrm{row}} \times N_{\mathrm{cent}}^{\mathrm{col}}$	normalized co-located 6×6 UPA, dual-polarized `VH`
UE array topology $N_{\mathrm{UE}}^{\mathrm{row}} \times N_{\mathrm{UE}}^{\mathrm{col}}$	1×1 planar array, dual-polarized `VH`
Element spacing Δ_v, Δ_h	0.5λ vertical, 0.5λ horizontal
Element radiation pattern	3GPP `tr38901`
User height h_{UE}	1.5 m
Number of users K	10
Simulation horizon T_{sim}	3600 s
Time step Δt	5 s
Distributed APs per strategy N_{mov}	6
Fixed distributed APs N_{fix}	0
Candidate wall height h_{cand}	1.5 m
Candidate wall spacing s_{wall}	18 m
Minimum candidate spacing $d_{\mathrm{cand}}^{\mathrm{min}}$	12 m
Candidate wall offset Δ_{wall}	0.25 m
Relocation window T_{rel}	60 s
Optimization distance threshold d_{th}	25 m

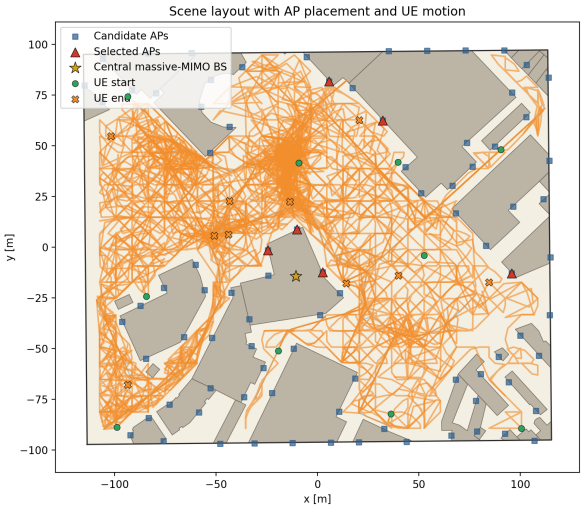


Fig. 2: Geometry results for the Rabot scenario. The panels show the selected deployment in the urban layout and the simulated user-trajectory overlay over the walkable area.

VI. Scenario Geometry and Mobility Results

This result block summarizes the simulated environment before presenting radio-performance metrics. It reports the available candidate mounting locations, the selected deployment, and the spatial distribution of user trajectories over the Rabot area.

Figure 2 provides the geometric context for the subsequent SINR, ESR, and relocation analyses. The scene-layout panel reports the candidate-site pool and the selected constellation, whereas the trajectory overlay reports the spatial spread of the simulated user population. For the present Rabot instance, the

wall-generation stage yields 87 candidate low-height mounting locations, from which 6 distributed sites are activated per distributed strategy. The simulated mobility trace contributes 7210 user-snapshot positions over the 1 h horizon, and these positions span most of the accessible outdoor footprint rather than a single narrow corridor. This geometry already suggests that a single rooftop proxy will face a harder coverage problem than wall-mounted distributed deployments, because the user population repeatedly moves through streets and courtyard edges that can be served from multiple nearby façades.

VII. SINR- and ESR-Oriented Results

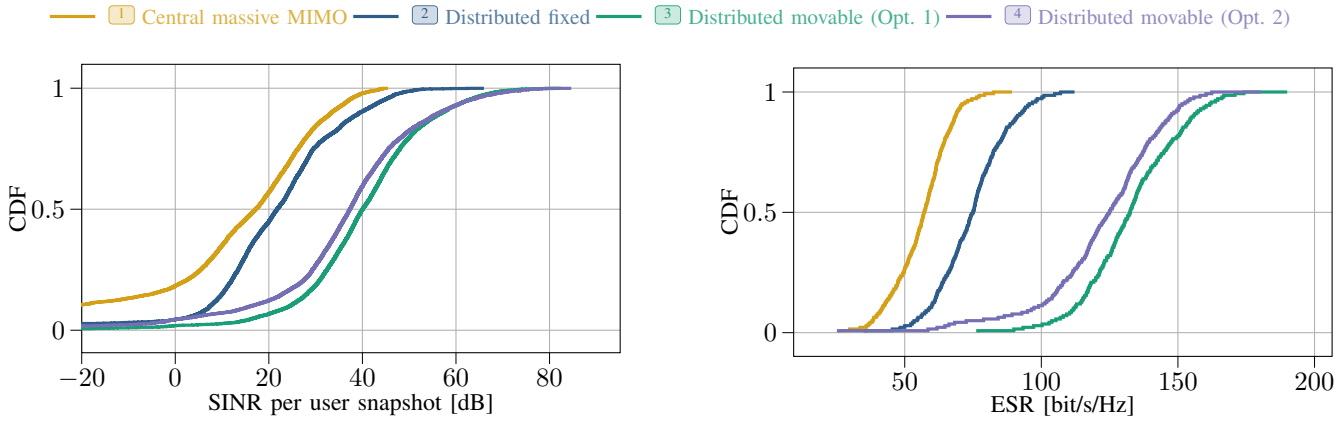
This section reports the radio-performance figures generated by the postprocessing workflow. The panels summarize global SINR statistics, user-level averages, temporal ESR evolution, aggregate ESR distributions, and worst-user behavior.

Figure 3 establishes a clear ordering. The nearest-user proxy-ESR controller of Distributed movable (Opt. 1) dominates almost the full empirical SINR range. Its mean SINR reaches 39.7 dB, with a 10th percentile of 24.5 dB; Distributed fixed reaches 21.3 dB mean and 7.5 dB at the 10th percentile. The outage fraction below 0 dB drops from 4.41 % for Distributed fixed to 1.87 % for Distributed movable (Opt. 1). Distributed movable (Opt. 2) is the second-best radio strategy with 35.7 dB mean SINR and 16.3 dB 10th-percentile SINR, but its outage fraction remains essentially identical to Distributed fixed at about 4.42 %. Hence, the power-only proxy improves the middle and upper portions of the distribution much more strongly than the weakest-link tail.

Central massive MIMO is weakest throughout, with 9.8 dB mean SINR, −22.1 dB 10th-percentile SINR, and 18.4 % outage. The per-user mean boxplot reinforces the same message: Distributed movable (Opt. 1) lifts all users into a narrow 36.4 dB to 42.5 dB band, Distributed movable (Opt. 2) places all users in a lower but still clearly elevated 31.1 dB to 38.8 dB band, whereas Central massive MIMO exhibits the widest user imbalance and even negative per-user means for the weakest users.

The ESR figures in Fig. 3 follow the same ranking. Mean ESR reaches 133.1 bit/s/Hz for Distributed movable (Opt. 1), 122.8 bit/s/Hz for Distributed movable (Opt. 2), 74.9 bit/s/Hz for Distributed fixed, and 56.4 bit/s/Hz for Central massive MIMO. Relative to Distributed fixed, Distributed movable (Opt. 1) improves the mean ESR by 77.8 % and the 10th-percentile ESR by 88.8 %. Distributed movable (Opt. 2) still improves the mean ESR by 64.1 %, but remains consistently below Distributed movable (Opt. 1). Window by window, Distributed movable (Opt. 1) attains the highest mean ESR in 42 of the 61 relocation windows and exceeds Distributed fixed in all windows; Distributed movable (Opt. 2) leads in 15 windows and exceeds Distributed fixed in 57 windows. The relocation markers every 60 s make the controller action visible as a sequence of step changes, and the highest sustained post-relocation levels occur for Distributed movable (Opt. 1).

Together, these figures show that the best result in this scenario is not merely “more movement” or “more power”, but the combination of relocation with a phase-aware UE-to-UE proxy-CSI objective. Distributed fixed remains a strong zero-relocation benchmark, whereas Central massive MIMO is clearly mismatched to the spatially spread Rabot user process.



(a) Empirical SINR distribution across 7210 user snapshots: 10 scheduled users over 721 time samples.

(b) ESR cumulative distribution function (CDF) across the 721 evaluated time samples.

Fig. 3: Radio-performance distributions for the Rabot trace. Distributed movable (Opt. 1) produces the most favorable SINR and ESR distributions, with Distributed movable (Opt. 2) as the next-best movable strategy.

VIII. Relocation and Practicality Results

The relocation figures complement the radio-performance results by quantifying the physical reconfiguration burden associated with movable infrastructure. The scheduling outputs report how often the movable APs change position and how much displacement the resulting schedules induce.

Figure 4 quantifies the price paid for movable infrastructure. As expected, Central massive MIMO and Distributed fixed induce no physical motion. Among the proxy-based movable controllers, Distributed movable (Opt. 1) reaches a move fraction of 80.0 % and a total traveled distance of 14.2 km, with a mean per-move distance of 49.2 m. Distributed movable (Opt. 2) is the more aggressive controller, with 311 moves, a move fraction of 86.4 %, a total traveled distance of 18.6 km, and a mean move distance of 59.9 m.

This ranking is important when combined with the radio metrics from Section VII. Distributed movable (Opt. 1) not only outperforms Distributed movable (Opt. 2) in mean and lower-tail SINR/ESR, it also requires less motion: about 24 % less total relocation distance and a smaller move fraction. Thus, within the two nearest-user proxy controllers, the phase-aware proxy-ESR rule is strictly preferable to the power-only proxy rule in the present scenario. Distributed fixed therefore remains the practical zero-mobility reference, while Distributed movable (Opt. 1) defines the best radio-performance point among the movable strategies.

The relocation-distance CDF adds one further observation: the movable controllers do not differ only in how often they move, but also in how broadly they explore the candidate pool. Distributed movable (Opt. 1) uses 140 distinct positions over the run, while Distributed movable (Opt. 2) uses 162. Hence, the power-only rule is not merely more active; it is also spatially less conservative. In this dataset, that additional exploration does not translate into better radio performance.

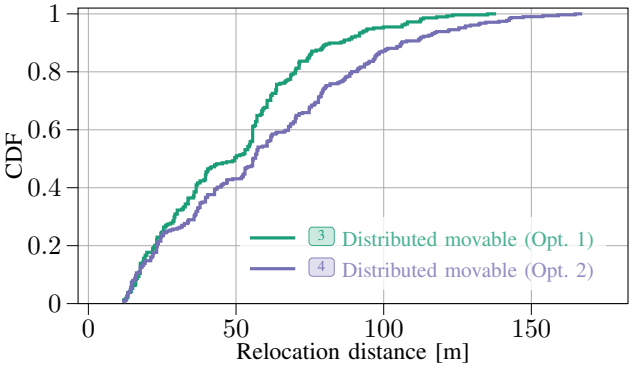


Fig. 4: Relocation-distance CDF used to interpret how much physical infrastructure movement is required by the compared placement strategies. Distributed movable (Opt. 1) achieves the strongest radio performance with a lower movement burden than Distributed movable (Opt. 2), whereas the fixed and central references remain static.

IX. Discussion and Current Limits

The present study has several modeling and evaluation limits that define the scope of the reported results.

Two findings stand out from the Rabot results. First, distributed wall-mounted deployments are consistently superior to the normalized central rooftop proxy in this scenario. Even Distributed fixed improves both average and lower-tail performance over Central massive MIMO, which indicates that the spatially spread outdoor user process benefits strongly from nearby façade-mounted service points. Second, the quality of the relocation proxy matters more than relocation alone. Distributed movable (Opt. 1) outperforms Distributed movable (Opt. 2) on all reported radio metrics while also requiring less movement, which shows that phase-aware UE-to-UE proxy CSI carries useful information beyond power-only proxy measurements.

The main limitations are the following:

- Only a single outdoor scenario is considered.

- The central massive-multiple-input multiple-output (MIMO) comparison point is a rooftop proxy, not a full placement model for one true co-located massive-MIMO array.
- The movable optimization results use trace-driven current-window look-ahead in the UE-to-UE proxy measurements. They should therefore be read as a window-level relocation benchmark, not as a strictly causal online controller.
- The relocation analysis reports aggregate movement burden rather than attributing marginal ESR gain to individual moves; this would require counterfactual no-move comparisons at each relocation instant.
- The current Rabot configuration disables coverage-map exports, so the paper centers on trajectory-level SINR and schedule metrics.
- Citizen participation is abstracted as movable candidate positions and scheduler actions rather than being validated experimentally.
- Sensitivity to random seeds, scenario perturbations, and broader deployment-cost models is not yet characterized.

These limitations motivate extensions toward multi-scenario studies, sensitivity analyses, coverage-map reporting, and eventual experimental validation in co-created deployments.

X. Conclusion

This article presented a simulation-based study of low-height distributed-MIMO access networks with repositionable infrastructure in an outdoor urban scenario. The paper combined a system model, two proxy-based relocation algorithms, a reproducible Rabot simulation workflow, and postprocessed geometry, SINR, ESR, and relocation figures derived from exported artifacts. The movable strategies span a nearest-user UE-to-UE proxy-CSI controller with a proxy-ESR objective and a nearest-user proxy-power controller that ranks subsets from stored UE-to-UE link-power samples scaled to the AP transmit-power budget.

For the Rabot dataset, Distributed movable (Opt. 1) is the best overall strategy. It yields the highest mean and lower-tail SINR/ESR, reduces outage most strongly, and does so with less movement than Distributed movable (Opt. 2). Distributed movable (Opt. 2) remains clearly beneficial relative to Distributed fixed in mean-rate terms, but it does not improve the outage tail and requires the largest relocation effort. Distributed fixed remains the strongest zero-relocation option, while Central massive MIMO is the weakest reference in this scenario.

Within its current scope, the framework supports systematic comparison of a central rooftop proxy, a fixed distributed deployment, and two relocation-capable distributed deployments under a common channel model and mobility process. Natural next steps are multi-scenario studies, coverage-map analysis, sensitivity analysis for the proxy-based relocation rules, and experimental validation in the broader Cocoon program.

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